

THE BEST DISSERTATION YOU WILL EVER READ
APACHE POINT OBSERVATORY FOLLOW-UP OF
ACCELERATING CANDIDATE EXOPLANET HOST STARS (APO
ACCELS)
AND CHARACTERIZATION OF THE BINARY HOST OF THE DIRECTLY
IMAGED PLANET HD 143811 AB b

by

ANNE ELISABETH PECK, B.A., M.S.

A dissertation submitted to the Graduate School

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Harrison E. Cook

Candidate

Astronomy

Major

This Dissertation is approved on behalf of the faculty of New Mexico State University, and is acceptable in quality and form for publication:

Approved by the Dissertation Committee:

Wladimir Lyra

Chairperson

Moire Prescott

Committee Member

Kristian Finlator

Committee Member

Michael Engelhardt

Committee Member

DEDICATION

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First and foremost, I want to thank and acknowledge my advisor Dr. Wladimir Lyra. Wlad, I am so appreciative for all that you've done for me. From the start, you took me on as your first Ph.D. student even though I said I was interested in AGN and galaxies not planets. After asking around, you were able to connect me with the AGN channel folks in New York and we embarked on a quick project to blow up a star in a disk that would only take seven years! On top of that, you nominated me for awards and scholarships practically looking under every planetesimal, pebble, and dust grain to find funding for me and my decidedly NOT planets research, though maybe there ARE planets in AGN disks. I'm also grateful for your support beyond the research. It got tough there for a while, and I'm uncertain I would have made it through without some of our breakfasts. Also, a huge thanks to you and Adrienn for hosting all the delicious Brazilian barbeques and dinner parties for the group throughout the years.

I would also like to express thanks to my committee for their part in getting me here. Dr. Moire Prescott, thanks for all of observational context that built my intuition for galaxies and AGN in the Galaxies I class. I'll always remember when

I mentioned in passing to Dr. Kristian Finlator that I had managed to successfully compile my supernova model on the Stampede 2 high performance computer using multiple nodes for the first time. Your simple “Congratulations! That’s a great accomplishment.” was unexpected and baffled me in the moment, since at the time it felt I had just done the barest of minimums, but it was most welcome and a compliment I didn’t realize I needed to hear. Dr. Michael Engelhardt, the general relativity will likely finally come in hand in the next phases of this work.

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My journey here started in 2012 while posted as a Marine Security Guard at the U.S. Embassy in Asunción, Paraguay. I had become fascinated with popular astronomy videos and started talking about them with everyone who would (and wouldn’t) listen. It was a small community, and eventually Ambassador James Thessin and Marcia Thessin caught wind. Marcia came by Post One while I was on duty one day and asked if I wanted to come to the private Ambassador’s quarters to have a video chat with their daughter Rachel Thessin who was an aerospace engineer and son-in-law Will Farr who was an astrophysicist to talk about getting into astronomy as a career. I was humbled and flattered that they would invite a lowly Marine for such a thing and readily agreed. At the appointed time, I arrived at the residence where Marcia walked me through the official quarters into the private residence and the small office where we connected with Rachel and Will. They taught me about how to find papers on arXiv.org, look for schools with astronomy degrees, and focus my community college courses in preparation for mathematics and physics. I had no idea how valuable that call would end

up being, but I took their advice to heart and immersed myself in astronomy magazines, podcasts, and documentaries. I even took pre-calculus for the second time at my next post in Kabul, Afghanistan where I had to travel in an up-armored vehicle while kitted up to another base in order to take proctored exams. Perhaps needless to say, my mind wasn't all in it, and I had to take pre-calculus a third time in my life when I got to Columbia University. But I passed and took accelerated calculus courses in order to start physics in my second year, since I didn't want to spend longer than four years in college. So, thank you all James, Marcia, and Rachel Thessin and Will Farr for the roles each of you played in getting me started on my journey to astronomy back in Asunción, Paraguay. I've dedicated much of my efforts to paying this forward.

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VITA

EDUCATION

2016 - 2020 B.A., Astronomy with High Honors
Smith College
Northampton, MA, USA 2021 - 2026
M.S., Astronomy
New Mexico State University
Las Cruces, NM, USA

FIELD OF STUDY

Major Field: High Energy Gravitational and Electromagnetic Wave Transients

PUBLICATIONS

Peck, A., Roberson, W., Nielsen, E., De Rosa, R., Jones, N., Wang, J., Macintosh, B., Lewis, B., Duchêne, G., Metchev, S., Abbas, A., Xuan, J., Sanghi, A., Patience, J., Barman, T., Bulger, J., Chilcote, J., Esposito, T., Fitzgerald, M., Follette, K., Gallamore, H., Goodsell, S., Graham, J., Greenbaum, A., Hiben, P., Ingraham, P., Kalas, P., Konopacky, Q., Marchis, F., Maire, J., Marois, C., Matthews, B., Mawet, D., Millar-Blanchaer, M., Oppenheimer, R., Palmer, D., Perrin, M., Poyneer, L., Pueyo, L., Rajan, A., Rameau, J., Rantakyö, F., Ren, B., Ruffio, J.B., Savransky, D., Schneider, A., Sivaramakrishnan, A., Smith, A., Song, I., Soummer, R., Thomas, A., Ward-Duong, K., Wolff, S., *Characterization of the Host Binary of the Directly Imaged Exoplanet HD 143811 AB b*. ApJ Letters, 995, 40, doi: 10.3847/2041-8213/ae1a67

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LEADERSHIP AND PROFESSIONAL EXPERIENCE

ABSTRACT

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Dr. Eric Nielsen, Advisor

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LIST OF ABBREVIATIONS

Instruments, Telescopes & Observatories

Fermi	Fermi Gamma-ray Space Telescope
INTEGRAL	International Gamma-ray Astrophysics Laboratory
KAGRA	Kamioka Gravitational Wave Detector
LIGO	Laser Interferrometer Gravitational Wave Observatory
LISA	Laser Interferrometer Space Antenna
LVK	LIGO-Virgo-KAGRA
Virgo	Virgo interferometer
ZTF	Zwicky Transient Facility

Acronyms and Jargon

1g, 2g, 3g	first, second, third generation black hole
AGN	active galactic nucleus
BBH	binary black hole
bf	bound-free
BH	black hole
BLR	broad line region
CO	compact object
EM	electromagnetic
EMRI	extreme-mass ratio inspriial
ff	free-free
GRB	gamma-ray burst
GW	gravitational wave
HLLE	Harten-Lax-van Leer-Einfeldt solver
IMBH	intermediate mass black hole
IMF	initial mass function
IMRI	intermediate-mass ratio inspiral
ISCO	inner-most stable circular orbit
LSST	Legacy Survey of Space and Time
McFACTS	Monte Carlo for AGN Channel Testing and Simulations
NSC	nuclear star cluster
PDS	pressure-driven snowplow
PISN	pair-instability supernova
O3, O4, O5	LVK observing runs 3, 4, and 5
SG	Sirko and Goodman disk model
SMBH	supermassive black hole
SLSN	superluminous supernova
SN	supernova
TQM	Thompson, Quataert, and Murray disk model

LIST OF SYMBOLS

Symbol	Meaning
$q = M_2/M_1, M_1 > M_2$	mass ratio
M_1	mass of more massive binary component
M_2	mass of less massive binary component
χ_{eff}	effective spin
τ_{AGN}	AGN lifetime
$R_{\text{g}} = \frac{GM}{c^2}$	gravitational radius
G	gravitational constant
M	mass
c	speed of light
$\chi_{1,2}$	dimensionless spin vectors
L_{bin}	binary orbital angular momentum
L_{disk}	disk angular momentum
L_{BH}	black hole orbital angular momentum
γ	index of the mass power-law distribution
$M_{\odot}$	solar mass
M_{BH}	supermassive black hole mass
μ_{χ}	mean spin
σ_{χ}	spin standard deviation
R_{out}	disk outer radius
H	disk scale height
e	orbital eccentricity
$e_{0,\text{max}}$	maximum initial eccentricity
e_{crit}	eccentricity threshold where a can evolve
a	orbital semi-major axis
$R_{\text{H}} = a_1(M_1/3M_{\text{BH}})^{1/3}$	Hill radius
M_{NSC}	nuclear star cluster mass
N_{BH}	number of black holes
$N_{\star}$	number of stars
R_{crit}	break radius in the NSC number density profile
η	power-law index for NSC profiles
n	number density
f_{ret}	fraction of retrograde binaries

Symbol	Meaning	<i>continued</i>
$\mathcal{R}$	BBH merger rate	
ε_{SN}	SN energy density	
$P_D(r)$	disk gas pressure at radius r	
T	gas temperature	
k	Boltzmann's constant	
$\mathcal{C}$	SN breakout parameter	
ρ	gas volume density	
ρ_0	disk midplane density	
E_{SN}	SN total kinetic energy	
R_{SN}	SN shock radius	
m_{p}	proton mass	
Ω_0	Keplerian orbital frequency	
q_{shear}	logarithmic shearing parameter	
E	total energy density	
c_{s}	sound speed	
ρ_{ej}	SN ejecta density	
M_{ej}	SN ejecta mass	
t_{PDS}	PDS transition time	
R_{PDS}	PDS transition radius	
f_{K}	kinetic energy fraction	
R_{m}	muffling radius	
R_{s}	Schwarzschild radius	
$T(z)$	vertical temperature profile	
T_{mid}	disk midplane temperature	
T_{iso}	isothermal temperature cap	
T_{floor}	minimum temperature floor	
T_{eff}	disk effective temperature	
H_T	temperature scale height	
τ	optical depth	
$\theta = 5040/T$	thermal parameter	
κ_{λ}	wavelength-dependent absorption opacity	
σ_{λ}	wavelength-dependent scattering opacity	
I_{λ}	specific intensity	

Symbol	Meaning	<i>continued</i>
S_λ	source function	
B_λ	Planck function	
J_λ	mean intensity	
ω_λ	single-scattering albedo	
I^+	upward intensity	
I^-	downward intensity	
$U = (I^+ + I^-)/2$	mean intensity (Feautrier)	
$V = (I^+ - I^-)/2$	flux (Feautrier)	
α_λ	total extinction coefficient	
F_λ	flux density	
H^-	negative hydrogen ion	
H^+	ionized hydrogen	
n_e	electron number density	
n_I	ion number density	
g_{ff}	free-free Gaunt factor	
g_{bf}	bound-free Gaunt factor	
P_e	partial electron pressure	
χ_n	excitation potential for level n	
$I = 13.598 \text{ eV}$	hydrogen ionization energy	
σ_0	absorption cross-section normalization	
λ	wavelength	
ν	frequency	
h	Planck constant (distinct from disk scale height H)	
m_e	electron mass	
χ_p	in-plane spin component	
$\mathcal{M}_{\text{ch}}$	binary chirp mass	
μ_{BBH}	BBH reduced mass	
D	distance to source	
ν_{GW}	gravitational wave frequency	
h_{char}	characteristic GW strain	
h_{str}	GW strain	
a_{BBH}	BBH semi-major axis	
M_{BBH}	BBH total mass	

Symbol	Meaning	<i>continued</i>
$\dot{\nu}_{\text{GW}}$	GW frequency time derivative	
N	number of GW cycles	
t_{damp}	orbital eccentricity damping timescale	
$h = H/r$	disk aspect ratio (distinct from Planck constant h)	
n_{AGN}	AGN number density	
α	disk viscosity parameter	
$p(R)$	probability density over radius	
ΔR	thickness of a spherical shell	

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xcolor

float wasysym xcolor rotating longtable multirow

Apache Point Observatory follow-up of ACcelerating Candidate ExopLanet
host Stars

(APO ACCELS): Ages for 166 Accelerating Stars in the Northern Hemisphere

[0000-0003-2461-6881]Anne E. Peck Department of Astronomy, New Mexico
State University, P.O. Box 30001, MSC 4500, Las Cruces, NM 88003, USA

[0000-0001-6975-9056]Eric L. Nielsen Department of Astronomy, New Mexico
State University, P.O. Box 30001, MSC 4500, Las Cruces, NM 88003, USA

[0000-0002-4918-0247]Robert J. De Rosa European Southern Observatory,
Alonso de Córdova 3107, Vitacura, Santiago, Chile

[0000-0001-5684-4593]William Thompson National Research Council, Herzberg
Astronomy and Astrophysics, Victoria, British Columbia

[0000-0003-1212-7538]Bruce Macintosh Department of Astronomy and Astro-
physics, UC Santa Cruz, Santa Cruz CA 95064

[0009-0008-9687-1877]William Roberson Department of Astronomy, New Mex-
ico State University, P.O. Box 30001, MSC 4500, Las Cruces, NM 88003, USA

[0000-0002-9156-9651]Adam J. R. W. Smith Department of Astronomy, New Mexico State University, P.O. Box 30001, MSC 4500, Las Cruces, NM 88003, USA

[0000-0003-3906-9518]Jessica Klusmeyer Department of Astronomy, New Mexico State University, P.O. Box 30001, MSC 4500, Las Cruces, NM 88003, USA

[0009-0006-4626-832X]Asif Abbas Department of Astronomy, New Mexico State University, P.O. Box 30001, MSC 4500, Las Cruces, NM 88003, USA

[0000-0001-9659-7486]Jason Jackiewicz Department of Astronomy, New Mexico State University, P.O. Box 30001, MSC 4500, Las Cruces, NM 88003, USA

[0000-0002-9771-9622]Jon Holtzman Department of Astronomy, New Mexico State University, P.O. Box 30001, MSC 4500, Las Cruces, NM 88003, USA

[0009-0000-8603-169X]Hannah Gallamore Department of Astronomy, New Mexico State University, P.O. Box 30001, MSC 4500, Las Cruces, NM 88003, USA

[0000-0002-0457-2941]Marah Brinjikji School of Earth and Space Exploration, Arizona State University, Tempe, AZ 85287, USA

Jennifer Patience School of Earth and Space Exploration, Arizona State University, Tempe, AZ 85287, USA

[0000-0002-9242-9052]Jayke S. Nguyen Department of Astronomy & Astrophysics, UC San Diego, La Jolla, CA 92093, USA

[0000-0001-7443-6550]Alex Madurowicz Department of Physics, Stanford University, Stanford, CA 94305, USA

[0000-0002-8711-7206]Dmitry Savransky Sibley School of Mechanical and Aerospace Engineering, Cornell University, Ithaca, NY 14853, USA

1. Introduction

2. Sample, observations, and data reduction

3. Lithium

4. Calcium Emission

5. Gyrochronology

6. Combined Age Posteriors

7. Discussion

8. Conclusions

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The *TESS* data products used in this work are available in the MAST archive, specifically the “*TESS* Light Curves - All Sectors” repository (?), the “*TESS* “Fast” Light Curves - All Sectors” repository (?), the “*TESS* Target Pixel Files - All Sectors” repository (?), and the “*TESS* “FAST” Target Pixel Files - All Sectors” repository (?).

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